

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12399189
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Cho; Young Jun et al.

Immunoassay device and immunoassay method

Abstract

The present invention relates to an immunoassay device and an immunoassay method. According to an aspect of the present invention, an immunoassay device includes a measurement unit provided with a detection unit disposed at one side of a stage accommodating cartridges having a plurality of wells to move in a direction, in which the plurality of cartridges are arranged, and capable of measuring a state within the well disposed at the outermost side, and including a shielding plate that moves to cover an opened upper portion of the well disposed at the outermost side to block introduction of light into the well.

Inventors:	Cho; Young Jun (Daejeon, KR), Oh; Jae Hoon (Daejeon, KR), Park; Moon Kyoo (Daejeon, KR), Lee; Jeong Wook (Daejeon, KR)
Applicant:	INVITROS CO., LTD. (Seoul, KR)
Family ID:	1000008780534
Assignee:	INVITROS CO., LTD. (Seoul, KR)
Appl. No.:	17/598644
Filed (or PCT Filed):	March 20, 2020
PCT No.:	PCT/KR2020/003907
PCT Pub. No.:	WO2020/197196
PCT Pub. Date:	October 01, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20220163548 A1	May. 26, 2022

Foreign Application Priority Data

Publication Classification

Int. Cl.: **G01N35/00** (20060101); **G01N33/53** (20060101); **G01N35/02** (20060101); **G01N35/04** (20060101); **G01N35/10** (20060101)

U.S. Cl.:

CPC **G01N35/00029** (20130101); **G01N33/5302** (20130101); **G01N35/028** (20130101); **G01N35/1009** (20130101); G01N2035/00099 (20130101); G01N2035/0418 (20130101)

Field of Classification Search

CPC: G01N (35/00029); G01N (33/5302); G01N (35/028); G01N (35/1009); G01N (2035/00099); G01N (2035/0418)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5158895	12/1991	Ashihara et al.	N/A	N/A
5290708	12/1993	Ashihara et al.	N/A	N/A
5401465	12/1994	Smethers et al.	N/A	N/A
5482839	12/1995	Ashihara et al.	N/A	N/A
5682232	12/1996	Tajima et al.	N/A	N/A
5773305	12/1997	Zabetakis et al.	N/A	N/A
6509193	12/2002	Tajima	N/A	N/A
7452508	12/2007	Jacobs et al.	N/A	N/A
2001/0007770	12/2000	Tajima	N/A	N/A
2002/0123156	12/2001	Tajima	N/A	N/A
2005/0282182	12/2004	Tajima et al.	N/A	N/A
2006/0197955	12/2005	Koike	N/A	N/A
2010/0129789	12/2009	Self et al.	N/A	N/A
2010/0278698	12/2009	Tajima	N/A	N/A
2011/0009608	12/2010	Kim et al.	N/A	N/A
2011/0262919	12/2010	Tajima	N/A	N/A
2012/0122231	12/2011	Tajima	N/A	N/A
2012/0135394	12/2011	Kim et al.	N/A	N/A
2012/0140231	12/2011	Knox et al.	N/A	N/A
2012/0195811	12/2011	Nelson	N/A	N/A
2013/0121881	12/2012	Ishizawa et al.	N/A	N/A
2014/0048540	12/2013	Tajima	N/A	N/A
2014/0284223	12/2013	Malecha et al.	N/A	N/A
2015/0266658	12/2014	Tajima	N/A	N/A
2015/0309059	12/2014	Tajima	N/A	N/A
2016/0025722	12/2015	Tajima	N/A	N/A
2016/0069921	12/2015	Holmes et al.	N/A	N/A
2016/0169880	12/2015	Holmes et al.	N/A	N/A

2016/0245756	12/2015	Tajima	N/A	N/A
2017/0219615	12/2016	Matsumoto et al.	N/A	N/A
2017/0269110	12/2016	Horak	N/A	N/A
2017/0269114	12/2016	Bryant et al.	N/A	N/A
2019/0086436	12/2018	Lachance	N/A	N/A
2021/0132097	12/2020	Bryant et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102246043	12/2010	CN	N/A
107024593	12/2016	CN	N/A
206638610	12/2016	CN	N/A
109070084	12/2017	CN	N/A
0725929	12/1995	EP	N/A
0763739	12/2004	EP	N/A
2383575	12/2010	EP	N/A
2439537	12/2011	EP	N/A
2600156	12/2012	EP	N/A
2921861	12/2014	EP	N/A
H09-061341	12/1996	JP	N/A
H10-197449	12/1997	JP	N/A
2002-055050	12/2001	JP	N/A
3627593	12/2004	JP	N/A
3682302	12/2004	JP	N/A
3705055	12/2004	JP	N/A
3721889	12/2004	JP	N/A
2008-170332	12/2007	JP	N/A
2008-175708	12/2007	JP	N/A
2010-060537	12/2009	JP	N/A
5143570	12/2012	JP	N/A
2014-508034	12/2013	JP	N/A
5878254	12/2015	JP	N/A
10-1995-0014745	12/1994	KR	N/A
10-0148239	12/1997	KR	N/A
10-2005-0113604	12/2004	KR	N/A
10-2011-0106892	12/2010	KR	N/A
10-1307978	12/2012	KR	N/A
10-2014-0031200	12/2013	KR	N/A
10-1423936	12/2013	KR	N/A
10-2015-0026003	12/2014	KR	N/A
494239	12/2001	TW	N/A
I461690	12/2013	TW	N/A
I600891	12/2016	TW	N/A
I624543	12/2017	TW	N/A
I625521	12/2017	TW	N/A
2015-053290	12/2016	WO	N/A
2017-0161053	12/2016	WO	N/A

OTHER PUBLICATIONS

JPH1019449A, English translation attached. (Year: 1997). cited by examiner
Examination Report issued on Oct. 11, 2023, for the European patent application No. 20776679.1, 8 pages. cited by applicant
The Extended Search Report for EP application No. 20776679.1 dated Mar. 9, 2022, (10 pages). cited by applicant
The Extended Search Report for EP application No. 20778112.1 dated Mar. 11, 2022, (8 pages). cited by applicant
Extended European Search Report dated Jun. 29, 2022, issued in the corresponding European Patent Application No. 20776353.3, 11 pages. cited by applicant
International Search Report dated Jun. 24, 2020, issued in the corresponding International Application No. PCT/KR2020/003907, 4 pages. cited by applicant

Primary Examiner: Emch; Gregory S

Assistant Examiner: Dunn; McKenzie A

Attorney, Agent or Firm: ROTHWELL, FIGG, ERNST & MANBECK, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is a 35 U.S.C. 371 National Phase Entry Application from PCT/KR2020/003907, filed on Mar. 20, 2020 and designating the United States, claims the benefit of the priority of Korean Patent Application No. 10-2019-0036150, filed on Mar. 28, 2019, all of which are hereby incorporated by references in their entirety.

TECHNICAL FIELD

(2) The present invention relates to an immunoassay device and an immunoassay method.

BACKGROUND OF THE INVENTION

(3) As medical and biotechnology technologies advance, tests for detecting various molecular indicators such as blood cells, genes, proteins, antigens, and pathogens in biological samples such as blood and urine are being conducted. An inspection process is generally performed by taking a sample and analyzing and observing a change occurring after allowing the collected sample with a predetermined reagent that is suitable for a desired indicator.

(4) One of the techniques that are widely used in such an inspection process is an immunoassay method based on specific binding between antigens/antibodies.

(5) The immunoassay method may be classified into a radioimmunoassay (RIA), in which a signal is detected using radioactive isotopes, an enzyme-linked immunosorbent assay (ELISA) or enzyme immunoassay (EIA) that uses signal amplification by enzymes, a fluorescence antibody technique (FA) using fluorescence, chemiluminescence immunoassay (CLIA) using chemiluminescence, and the like, according to a detection principle and method thereof. In addition, the immunoassay method may be variously classified according to a method of using a marking material or types of substrates.

(6) An immunoassay device for implementing the immunoassay method according to the related art has a disadvantage that it is inefficient to test various samples because a cartridge having the same reaction method is disposed on one stage.

(7) In addition, the immunoassay device according to the related art has a disadvantage that it takes a long time to measure a state of a final reaction solution.

(8) Also, in the immunoassay device according to the related art, magnetic particles are discharged

to the outside of a tip to cause a risk of incorrect immunity testing.

BRIEF SUMMARY OF THE INVENTION

(9) Embodiments of the present invention are proposed to solve the above problems, and the present invention is to provide an immunoassay device and an immunoassay method, which are capable of performing various reaction methods on one state.

(10) In addition, the present invention is to provide an immunoassay device and an immunoassay method which are capable of reducing a time taken to measure a state of a final reaction solution.

(11) In addition, the present invention is to provide an immunoassay device and an immunoassay method, which are capable of performing a more precise reaction inspection because magnetic particles are held in a tip a process of implementing a reaction method.

(12) In addition, the present invention is to provide an immunoassay device and an immunoassay method, in which reaction and inspection of a sample and a reagent occurs in one small device.

(13) An immunoassay device according to an embodiment of the present invention includes: a stage which is capable of accommodating a plurality of cartridges having a plurality of wells that are opened upward and capable of surrounding a circumference of the well disposed at the outermost side of the cartridges; a solution transfer unit including a plurality of tips, which are movable relative to the stages, are disposed to correspond to positions of the cartridge, and suction a solution stored in the well or discharge the suctioned solution from the well; and a measurement unit disposed at one side of the stages to move in a direction in which the plurality of cartridges are arranged, provided with a detection unit that is capable of measuring a state within the well disposed at the outermost side, and including a shielding plate that moves to cover an opened upper portion of the well disposed at the outermost side to block introduction of light into the well.

(14) In the immunoassay device according to an embodiment of the present invention, a first hole may be provided in one side surface of the stage that surrounds the circumference of the well, which is disposed at the outermost side, of the cartridges so as to measure a state of the solution, the detection unit may be provided on one side surface of the measurement unit; and when the detection unit is disposed on the side surface of the well disposed at the outermost side to measure the state of the solution, the light incident into the well through the first hole may be blocked.

(15) In the immunoassay device according to an embodiment of the present invention, a second hole through which an end of the tip is introduced into an upper portion of the well disposed at the outermost side may be provided in the shielding plate, the second hole may have a size less than that of an upper hole of the well disposed at the outermost side, and when the tip is introduced into the well disposed at the outermost side, the light incident into the well through the second hole may be blocked.

(16) In the immunoassay device according to an embodiment of the present invention, the detection unit and a second hole formed in the shielding plate may be provided in plurality to measure states of the solutions stored in the plurality of wells disposed at the outermost side at the same time.

(17) In the immunoassay device according to an embodiment of the present invention, an injection of the solution from the tip into the well disposed at the outermost side and the measurement of the state of the solution by the detection unit may be performed at the same time.

(18) In the immunoassay device according to an embodiment of the present invention, as the solution stored in the tip is injected into the well disposed at the outermost side, a luminant of the solution may emit light, and the detection unit may measure the emitted light.

(19) In the immunoassay device according to an embodiment of the present invention, a light emission time of the luminant may be within 10 seconds, and the measurement of the state of the solution by the detection unit may be performed within 10 seconds.

(20) In the immunoassay device according to an embodiment of the present invention, the solution transfer unit may include a magnetic force applying part that is capable of applying magnetic force toward the tip, and when the solution stored in the tip is injected into the well disposed at the outermost side, as the magnetic force applying part approaches the tip, magnetic particles may be

held inside the tip.

(21) The immunoassay device according to an embodiment of the present invention may further include a measurement unit driving part that is capable of driving the measurement unit in at least one direction, wherein the measurement unit may be reciprocated along the wells, which are disposed at the outermost side, of the plurality of cartridges.

(22) An immunoassay method according to an embodiment of the present invention includes: a step in which a tip moving along a cartridge and storing a solution containing magnetic particle conjugate is disposed above a well disposed at the outermost side; a step in which a measurement unit moves in one direction so as to be disposed on a side surface of the well disposed at the outermost side, and a shielding plate covers an upper portion of the well disposed at the outermost side; a step in which the solution stored in the tip is injected to the inside of the well; and a step of measuring a state of the solution by a detection unit while injecting the solution.

(23) In the immunoassay method according to an embodiment of the present invention, the step in which the measurement unit moves in the one direction so as to be disposed on the side surface of the well disposed at the outermost side may include a step of blocking light incident into the well through a first hole provided in one side surface of the stage, which is capable of surrounding a circumference of the well disposed at the outermost side.

(24) In the immunoassay method according to an embodiment of the present invention, the step in which the tip descends into the well disposed at the outermost side may include a step of blocking the light incident into the well through a hole formed in an upper portion of the well disposed at the outermost side.

Advantageous Effects

(25) In the immunoassay device and the immunoassay method according to the embodiments of the present invention, the various reaction methods may be performed on one stage.

(26) Also, the time taken to measure the state of the final reaction solution may be reduced.

(27) Also, in the process of implementing the reaction method, the magnetic particles may be held in the tip to perform the more precise immunoassay.

(28) Also, the reaction and inspection of the sample and the reagent may occur in one small device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic perspective view illustrating a configuration of an immunoassay device according to an embodiment of the present invention.

(2) FIG. 2 is a schematic view illustrating a side surface of a solution transfer unit of FIG. 1.

(3) FIG. 3 is a schematic cross-sectional view illustrating a magnetic force applying part of FIG. 1.

(4) FIG. 4 is a schematic view illustrating an operation path of a measurement unit of FIG. 1.

(5) FIG. 5 is a view illustrating a front surface, a side surface, and a top surface of the measurement unit of FIG. 1.

(6) FIG. 6 is a view illustrating a measurement unit of FIG. 5 according to another embodiment.

(7) FIG. 7 is a block diagram of various driving units associated with a control unit of FIG. 1.

(8) FIG. 8 is a flowchart illustrating a process of measuring a state of a solution by the measurement unit of FIG. 1.

(9) FIG. 9 is a schematic view of a plurality of cartridges of FIG. 1 and a reagent stored in each of wells of the cartridges.

(10) FIG. 10 is a flowchart illustrating a process of a first step assay using a first cartridge of FIG. 9.

(11) FIG. 11 is a flowchart illustrating a process of a first step dilution assay using a second cartridge of FIG. 9.

(12) FIG. 12 is a flowchart illustrating a process of a second step assay using a third cartridge of FIG. 9.

(13) FIG. 13 is a flowchart illustrating a process of a second step dilution assay using a fourth cartridge of FIG. 9.

DETAILED DESCRIPTION OF THE INVENTION

(14) Hereinafter, specific embodiments of the present invention will be described in detail with reference to the drawings.

(15) Moreover, detailed descriptions related to well-known functions or configurations will be ruled out in order not to unnecessarily obscure subject matters of the present invention.

(16) FIG. 1 is a schematic perspective view illustrating a configuration of an immunoassay device according to an embodiment of the present invention, FIG. 2 is a schematic view illustrating a side surface of a solution transfer unit of FIG. 1, FIG. 3 is a schematic cross-sectional view illustrating a magnetic force applying part of FIG. 1, FIG. 4 is a schematic view illustrating an operation path of a measurement unit of FIG. 1, FIG. 5 is a view illustrating a front surface, a side surface, and a top surface of the measurement unit of FIG. 1, and FIG. 7 is a block diagram of various driving units associated with a control unit of FIG. 1.

(17) Referring to FIGS. 1 to 5 and 7, an immunoassay device **10** according to an embodiment of the present invention may include a stage **100** capable of accommodating a cartridge C, a solution transfer unit **200** including a plurality of tips **210** that is capable of suctioning a solution stored in the cartridge C or discharging the solution suctioned from the cartridge C, a measurement unit **300** measuring a state of the solution stored in the cartridge C, and a control unit **400** controlling the solution transfer unit so that the stage **100** and the solution transfer unit **200** move relative to each other, and at least a portion of the tips **210** suction the same content from the plurality of cartridges at the same time or discharges the same content to the plurality of cartridges at the same time. Here, the content may refer to a solution in the tip **210** or a solution in a well W in each of steps as the solution transfer unit **200** and the stages relatively move to perform each of the steps for reaction.

(18) The cartridge C according to an embodiment of the present invention stores reagents for detection and/or analysis of an analyte contained in a sample. The term “sample” used in this embodiment refers to a compound or composition to be analyzed, which contains the analyte, and the sample that is capable of being used in the present invention may be a liquid phase or liquid-like fluidic material. In an embodiment of the present invention, the sample may be a biological sample, i.e., be a bio-derived component such as whole blood, plasma, serum, urine, saliva, manure, and cell extract. In this embodiment, a left-right direction is described in a Y-axis direction, and a vertical direction is described in a Z-axis direction based on FIG. 2. An X-axis direction is described in a direction perpendicular to the Y-axis direction and the Z-axis direction. In addition, the directional setting may be merely an example, and the direction indicated by each axis direction may be set differently according to an embodiment.

(19) The stage **100** includes a cartridge accommodation part **110** that is capable of accommodating the plurality of cartridges C having the plurality of wells W, and a Y-axis guide part **120** that is capable of guiding movement of the cartridge accommodation part **110**.

(20) The cartridge accommodation part **110** may be provided in a planar shape extending in the X-axis and Y-axis directions and may include a plurality of grooves that are capable of accommodating the wells W of the cartridges.

(21) The grooves formed in the cartridge accommodation part **110** may be arranged in an M×N matrix. Here, M and N are arbitrary natural numbers, M rows are parallel to the Y-axis, and N columns are parallel to the X-axis. For example, when arranged in an 8×19 matrix as described in the embodiment of the present invention, the cartridge accommodation part **110** may include 152 grooves. However, the number of M rows and N columns is merely an example and thus is not limited thereto.

(22) One cartridge C may be accommodated in one row M of the cartridge accommodation part

110. For example, as described in the embodiment of the present invention, eight cartridges may be disposed in the cartridge accommodation part **100** having eight rows **M1** to **M8**.

(23) Reagents having different reaction methods may be stored in each of the cartridges **C**. Thus, various immunoassays may be performed on one stage **100**, and detailed description thereof will be described later.

(24) The cartridge accommodation part **110** may reciprocate in the Y-axis direction. As the cartridge accommodation part **110** in which the cartridge **C** is disposed moves in the Y-axis direction, and the solution transfer unit **200** to be described later moves in the Z-axis, the solution stored in the well **W** of the cartridge may be suctioned into the plurality of tips **210**, and the solution suctioned from the well **W** may be discharged.

(25) Also, a Y-axis driving part (not shown) that is capable of allowing the cartridge accommodation part **110** to move in the Y-axis direction may be provided.

(26) A lower portion of the cartridge accommodation part **110** may include a heating block (not shown) capable of being adjusted in temperature to cultivate the solution (a mixture of the sample and the reagent) stored in the well **W**. Here, the heating block may be disposed to surround a lower end of each of the wells **W**, and it is possible to adjust the temperature for the cultivation of the solution in the well.

(27) The Y-axis guide unit **120** may guide the cartridge accommodation part **110** so as to be reciprocated in the Y-axis direction. Also, the Y-axis guide unit **120** may surround the outside of the heating block and block external light from being introduced into the cartridge **C**.

(28) The solution transfer unit **200** may include the plurality of tips **210**, a punching part (not shown) disposed in front of the tips **210** to punch a film of the well **W**, and a magnetic force applying part **230** disposed behind the tips **210** to fix magnetic particles suctioned into the tips **210**. Here, the front side may be a direction in which the first well in the Y-axis direction is disposed.

(29) Also, the solution transfer unit **200** may include a driving part **240** capable of independently applying a pressure to each of the plurality of tips **210**, a Z-axis driving part **250** capable of allowing the plurality of tips **210** and the punching part to move in the Z direction, and a magnet driving part **260** capable of allowing the magnetic force applying part **230** to move in the Y-axis direction.

(30) The plurality of tips **210** may suction the solution stored in the well **W** or discharge the solution suctioned from the well **W**. For example, the plurality of tips **210** may be provided in eight (first tips **211** to eighth tips **218**), which respectively suction the stored in each of the wells **W** of the cartridges disposed in the first row **M1** to eight row **M8** or discharge the solution suctioned from the wells **W**.

(31) The plurality of tips **210** may be disposed in the same row to move along the same row. For example, the eight tips **211** to **218** may move along the same column **N1** to **N19** and then be introduced into the respective wells of the cartridge **C** disposed in the same column **N1** to **N19** to move. As described above, the plurality of tips **210** may move along the same column **N1** to **N19** to adjust an arrangement of the column of the wells **W** of each of the cartridges **C** so that a plurality of reaction methods are executed at the same time. This will be described below in detail.

(32) Also, each of the plurality of tips **210** may independently suction or discharge the solution by the driving part **240**, and detailed descriptions thereof will be described later. The plurality of tips **210** may be separated from the solution transfer unit **200** and may be mounted on the solution transfer unit **200** after the punching part punches each of the wells **W** of the cartridge **C**.

(33) The punching part (not shown) may be a constituent that punches the well **W** of the cartridge **C** to form a hole and may have an end having a pointed shape. Also, the punching part (not shown) may have a number corresponding to the number of cartridges **C**. For example, the punching part (not shown) may be provided in eight to correspond to the number of cartridges **C** disposed in the first to eight rows **M1** to **M8**.

(34) When the tip **210** is separated from the solution transfer unit **200**, the punching part may have

a length at which the only the punching part (not shown) reaches the film of the well W to punch the film, and when the tip **210** is mounted on the solution transfer unit **200**, the punching part may have a length at which the punching part (not shown) does not reach the well W. That is, the length of the tip **210** may be provided longer than that of the punching part (not shown).

(35) Also, the plurality of tips **210** and the punching part (not shown) may move in the vertical direction (Z-axis direction). Here, movement of the plurality of tips **210** and the punching part (not shown) in the Z-axis direction may be performed depending on each other, but is not limited thereto, and movement of the plurality of tips **210** and the punching part (not shown) in the Z-axis direction may be independent.

(36) The magnetic force applying part **230** may be disposed behind the plurality of tips **210** (a direction that is away from the first well of the cartridge in the Y-axis) to move toward the tips, thereby fixing the magnetic particles inside the tips **210**.

(37) The magnetic force applying part **230** may include a magnet **232** movable in a direction toward or away from the tip **210** from one side surface of the tip **210** and a magnet moving part **234** for allowing the magnet **232** to move in one direction.

(38) The magnet **232** may be a shape (a shape that is mutually fitted with) corresponding to a circumference of the tip **210**. For example, when the tip **210** is viewed from the top side, if the tip **210** has a circular shape, the magnet **232** may be a concave shape that is fitted with the tip (see FIG. 3).

(39) Since the magnet **232** and the tip **210** are fitted with respect to each other, magnetic force applied to the magnetic particles inside the tip **210** may be constantly adjusted. Also, when viewed from the top side of the tip **210**, the magnet **232** may wrap more than half around the tip **210**.

(40) Also, when viewed from one side of the tip **210**, a length of the magnet **232** (Z-axis direction) may be more than half of the length of the tip **210**. For example, the length of the magnet **232** may be 80% or more of the length of the tip **210**. Since the magnet **232** has the above-described length, the magnetic particles may be fixed on a large area inside the chip **210**.

(41) The magnet **232** may be integrally formed. Also, the magnets **232** may be provided in a plurality along the X-axis direction to independently move toward or be away from the plurality of tips **210**. For example, when the eight tips **211** to **218** are provided, eight magnets **232a** to **232h** corresponding to the eight tips **211** to **218** may be provided. In this case, eight magnet moving parts **234a** to **234h** capable of independently driving each of the magnets **232a** to **232h** may be provided.

(42) Also, the magnetic force applying part **230** may be an electromagnet, in which the magnetic force is changed in magnitude, instead of the permanent magnet. In this case, the magnetic particles may be fixed inside the tip **210** by changing only the magnitude of the magnetic force without moving toward the tip **210** or without using a moving device.

(43) A process in which the magnetic particles are held inside the tip **210** by the magnetic force applying part **230** will be described as follows.

(44) First, a fixture solution containing magnetic particles is introduced into the tip **210** by a suction pressure of the driving part **240**. Thereafter, a cleaning solution stored in the well W of the cartridge C is introduced into the tip **210**. Thereafter, a pneumatic pressure of the driving part **230** is adjusted to generate a flow inside the tip. Then, the magnetic force applying part **230** moves in the direction toward the tip **210** to fix the magnetic particles inside the tip **210**. Then, the solution containing impurities may be discharged to the well W, and thus, the magnetic particles may be fixed inside the tip. As described above, precision of the reaction may increase by holding the magnetic particles inside the tip **210**.

(45) The driving part **240** may independently apply a pressure to the plurality of tips **210**. For example, the driving part **240** may be a pump provided with pneumatic pressure.

(46) The driving part **240** may not only provide a pressure for allowing each of the tips **210** to suction or discharge the solution by using the pneumatic pressure, but also cause mixing (flow of the solution) of the solution inside the tip **210**. For example, the driving part **240** may adjust the

pressure inside the tip **210** by suctioning air inside the tip **210** or discharging air toward the inside of the tip **210**. Due to the change in pressure inside the tip **210**, the solution inside the tip **210** may flow in the vertical direction, and thus the mixing of the solution may occur inside the tip **210**.

(47) The driving part **240** may independently apply a pressure to each of the tips **210**. Particularly, when each of the tips **210** is introduced into the empty well of the cartridge C, the driving part **240** may not provide a pressure to the tip **210** introduced into the empty well. For example, when the first tip **211** is introduced into the well W**16** disposed in the sixth column N**6** of the first cartridge C**1**, the driving part **240** may provide a pressure for suctioning a conjugate solution stored in the well W**16**, and simultaneously, when the third tip **213** is introduced into the well W**36** disposed in the sixth column N**6** of the third cartridge C**3**, since no solution is stored in the well W**36**, a pressure may not be applied to the third tip **213**. This illustrates only two tips **211** and **213** of the eight tips **211** to **218**, and thus, the remaining tips **212** and **214** to **218** may operate in the same principle.

(48) The measurement unit **300** may be movable along one side (for example, the X-axis direction) of the stage **100** may measure a state of a solution from the tip **210** to any one well W of the cartridge **100**. Also, a measurement unit driving part (not shown) that is capable of allowing the measurement unit **300** to move in the X-axis direction may be provided.

(49) The measurement unit **300** may include a detection unit **310** that is capable of measuring an internal state of the well W disposed at the outermost side of each of the cartridges C and a shielding plate **320** that shields an opened upper portion of the well W disposed at the outermost side to block introduction of light into the well W.

(50) The measurement unit **300** may be movable along one side of the stage **100** and measure a state of the well disposed at the outermost side of each of the cartridges C. Also, the measurement unit **300** may be an optical reading module that optically analyzes a reaction result of the sample and the reagent.

(51) The measurement unit **300** may be disposed on a side surface of the well W disposed at the outermost side to complete the measurement in a short time when the solution is injected from the tip **210** to the well W disposed at the outermost side. For example, the measurement of the state of the solution by the measurement unit **300** may be performed within 20 seconds.

(52) Also, the measurement unit **300** of the present invention may be provided in a flash type, and a darkroom may be required to measure light emitted from a luminant of the solution.

(53) In this embodiment, for the condition of the darkroom, a circumference of the well W disposed at the outermost side of each of the cartridges C may be surrounded by a block (or the heating block), and also, the detection unit **310** and the shielding plate **320** may be used for the circumference of the well W.

(54) The detection unit **310** may measure a state within the solution of the well W disposed at the outermost side (for example, the detection unit **310** may be any known measurement unit such as a camera), and the detection unit may have an outer appearance that serves to block the introduction of light into the well W disposed at the outermost side.

(55) For example, a first hole H**1** may be provided in one side surface of the stage **100** capable of surrounding the circumference of the well W disposed at the outermost side of the cartridge C, and the detection unit **310** may be introduced into the first hole H**1** to block light introduced from the first hole H**1**.

(56) Here, a shape and size of the detection unit **310** may be provided to correspond to those of the first hole H**1**. For example, when the first hole H**1** has a rectangular shape, the shape of the detection unit **310** may also have a rectangular shape.

(57) The shielding plate **320** may function to block the introduction of light to an upper portion of the well W disposed at the outermost side. Also, the shielding plate **320** may extend from an upper edge of a main body **330** of the measurement unit to block the upper portion of the well W disposed at the outermost side. However, a second hole H**2** through which an end of the tip **210** is introduced

into the well W disposed at the outermost side may be provided in the shielding plate **320**.

(58) The second hole H2 may be formed less than an upper hole of the well W disposed at the outermost side, and when the tip **210** is introduced into the well W disposed at the outermost side, the light introduced into the well through the second hole H2 may be blocked.

(59) Also, when the detection unit **310** has a convex shape protruding in the Y-axis direction, the measurement unit **300** may move in the X-axis direction and be disposed in the well W disposed at the outermost side, and then move by the convex shape protruding in the Y-axis direction.

(60) The control unit **400** may control various driving parts for driving the invention. For example, the control unit **400** may control a Y-axis driving part (not shown) capable of allowing the cartridge accommodation part **110** to move in the Y-axis direction, a driving part **240** that is capable of independently applying a pressure to each of the plurality of tips **210**, a Z-axis driving part **250** capable of allowing the plurality of tips **210** and the punching part (not shown) to move in the Z-axis direction, a magnet driving part **230** capable of allowing the magnetic force applying part **230** to move in the Y-axis direction, and a measurement unit driving part (not shown) capable of allowing the measurement unit **300** to move in the X-axis direction and the Y-axis direction.

(61) The control unit **400** may control the tips so that at least the other tip stands by above the empty well while at least one tip of the plurality of tips suctions the solution stored in the well or discharge the suctioned solution from the well.

(62) FIG. **6** is a view illustrating a measurement unit of FIG. **5** according to another embodiment.

(63) Referring to FIG. **6**, a plurality of second holes H2 formed in the detection unit **310** and the shielding plate **320** may be provided to simultaneously measure internal states of the solutions of the plurality of wells disposed at the outermost side. Here, a distance between the plurality of detection unit **310** and the plurality of second holes H2 may be the same as a distance between a plurality of tips **210**.

(64) In this case, the solution injection from the tip **210** into the well W disposed at the outermost side and the measurement of the state of the solution of the detection unit **310** may occur at the same time.

(65) FIG. **8** is a flowchart illustrating a process of measuring a state of a solution by the measurement unit of FIG. **1**.

(66) Referring to FIG. **8**, a process of measuring an internal state of a well W disposed at the outermost side of each of cartridges C by a measurement unit **300** will be described as follows.

(67) First, while the cartridge moves, a tip **210** in which a solution containing a luminant, magnetic particles, and a pre-trigger is stored is disposed above the well W disposed at the outermost side (S1).

(68) Thereafter, a measurement unit **300** moves in the X-axis direction and is disposed on a side surface of a well W119 disposed at the first outermost side (S2). Here, light introduced from the side surface toward the well W119 disposed at the outermost side is blocked, and a shielding plate **320** covers an upper portion of the well W119 disposed at the outermost side.

(69) Thereafter, a tip **210** descends into the well W119 disposed at the outermost side (S3). Here, light introduced from an upper side toward the well W119 disposed at the outermost side is blocked (S3).

(70) Thereafter, a luminant of the solutions stored in the tip **210** is injected toward the well W119 disposed at the outermost side (S4). Here, a magnetic force applying part **230** approaches the tip **210**, and thus, the magnetic particles may be held inside the tip **210**.

(71) Thereafter, alternatively, a detection unit **310** may detect the luminant while injecting the solution to complete the measurement of the state of the solution injected from one tip **211** (S5).

(72) A measurement unit **300** may move to the X-axis, and then, the above-described process may be repeated to measure a state of solutions of the remaining wells W219 to W819 disposed at the outermost side.

(73) FIG. **9** is a schematic view of a plurality of cartridges of FIG. **1** and a reagent stored in each of

wells of the cartridges, FIG. 10 is a flowchart illustrating a process of a first step assay using a first cartridge of FIG. 9, FIG. 11 is a flowchart illustrating a process of a first step dilution assay using a second cartridge of FIG. 9, FIG. 12 is a flowchart illustrating a process of a second step assay using a third cartridge of FIG. 9, and FIG. 13 is a flowchart illustrating a process of a second step dilution assay using a fourth cartridge of FIG. 9.

(74) Referring to FIGS. 9 to 13, each of cartridges C disposed in each of rows M1 to M8 of a cartridge accommodation part 110 may store reagents having different reaction methods.

Accordingly, the cartridge C having a plurality of reaction methods may be disposed on one stage 100 to perform various immunoassays.

(75) For example, four cartridges C1 to C4 arranged in the rows M1 to M4 of the cartridge accommodation part 110 are disposed. A reagent for implementing first reaction is stored in the first cartridge C1, a reagent for implementing second reaction is stored in the second cartridge C2, a reagent for implementing third reaction is stored in the third cartridge C3, and a reagent for implementing fourth reaction is stored in the fourth cartridge C4. The plurality of reaction methods (the first to fourth reaction methods) may be implemented in a single procedure by relative movement of a solution transfer unit 200 and a cartridge accommodation part 110.

(76) In this embodiment, the first reaction, the second reaction, the third reaction, and the fourth reaction will be described as a first step assay, a first step dilution assay, a second step assay, and a second step dilution assay as examples, respectively. However, the first to fourth reactions are not limited thereto and may include all known reaction methods for the immunoassay.

(77) The first cartridge C1 is a container containing a reagent for implementing the first step assay, the second cartridge C2 is a container containing a reagent for implementing a first step dilution assay, the third cartridge C3 is a container containing a reagent for implementing the third step assay, and the fourth cartridge C4 is a container containing a reagent for implementing a fourth step dilution assay.

(78) A diluent that is capable of dilutes the sample, a conjugate solution containing a luminant, a fixture solution containing magnetic particles, a cleaning solution, and a pre-trigger that is capable of absorbing the luminant to provide side light may be stored in each of the wells W of the first to fourth cartridges C1 to C4. Also, each of the cartridges C1 to C4 may include an additional empty well W in addition to the empty well W for injecting the sample. Also, each of the cartridges C1 to C4 may optionally include a piercer rincer and a piercer cleaner, which are capable of removing foreign substances of a punching part (not shown).

(79) Also, the empty well W and the well W containing each of the reagents may be arranged so that the first step assay, the first step dilution assay, the second step assay, and the second step dilution assay are performed on the cartridge accommodation part 110 at the same time.

(80) For example, in the first cartridge C1, a well W11 of the first column N1 may be an empty well into which the sample is injected, the piercer rincer may be stored in a well W12 of the second column N2, the piercer cleaner may be stored in a well W13 of the third column N3, the conjugate solution may be stored in a well W16 of the sixth column N6, the fixture solution may be stored in a well W17 of the seventh column N7, the cleaning solution may be stored in wells W19, W110, W111, and W112 of the ninth column N9 to the twelfth column N12, the pre-trigger solution may be stored in a well W117 of the seventeenth column N17, a trigger solution may be stored in a well W119 of the nineteenth column N19, which is provided as the well for the side light, and wells W14, W15, W18, W113, W114, W115, W116, and W118 of the fourth column N4, the fifth column N5, the eighth column N8, the thirteenth column N13 to the sixteenth column N16, and the eighteenth column N18 may be empty wells.

(81) In the second cartridge C2, a well W21 of the first column N1 may be an empty well into which the sample is injected, the piercer rincer may be stored in a well W22 of the second column N2, the piercer cleaner may be stored in a well W23 of the third column N3, the diluent may be stored in a well W25 of the fifth column N5, the conjugate solution may be stored in a well W26 of

the sixth column N6, the fixture solution may be stored in a well W27 of the seventh column N7, the cleaning solution may be stored in wells W29, W210, W211, and W212 of the ninth column N9 to the twelfth column N12, the pre-trigger solution may be stored in a well W217 of the seventeenth column N17, the trigger solution may be stored in a well W219 of the nineteenth column N19, which is provided as the well for the side light, and wells W24, W28, W213, W214, W215, W216, and W218 of the fourth column N4, the eighth column N8, the thirteenth column N13 to the sixteenth column N16, and the eighteenth column N18 may be empty wells.

(82) In the third cartridge C3, a well W31 of the first column N1 may be an empty well into which the sample is injected, the piercer rincer may be stored in a well W32 of the second column N2, the piercer cleaner may be stored in a well W33 of the third column N3, the fixture solution may be stored in a well W37 of the seventh column N7, the conjugate solution stored in a well W38 of the eighth column N8, the cleaning solution may be stored in wells W39, W310, W311, W312, W313, W314, W315, and W316 of the ninth column N9 to the sixteenth column N16, the pre-trigger solution may be stored in a well W318 of the eighteenth column N18, the trigger solution may be stored in a well W319 of the nineteenth column N19, which is provided as the well for the side light, and wells W34, W35, W36, and W317 of the fourth column N4 to the sixth column N6 and the seventeenth column N17 may be empty wells.

(83) In the fourth cartridge C4, a well W41 of the first column N1 may be an empty well into which the sample is injected, the piercer rincer may be stored in a well W42 of the second column N2, the piercer cleaner may be stored in a well W43 of the third column N3, the diluent may be stored in a well W45 of the fifth column N5, the fixture solution may be stored in a well W47 of the seventh column N7, the conjugate solution stored in a well W48 of the eighth column N8, the cleaning solution may be stored in wells W49, W410, W411, W412, W413, W414, W415, and W416 of the ninth column N9 to the sixteenth column N16, the pre-trigger solution may be stored in a well W418 of the eighteenth column N18, the trigger solution may be stored in a well W419 of the nineteenth column N19, which is provided as the well for the side light, and wells W44, W46, and W417 of the fourth column N4, the sixth column N6, and the seventeenth column N17 may be empty wells.

(84) The above-described first to fourth cartridges C1 to C4 are disposed in the cartridge accommodation part 110, and the first step assay, the first step dilution assay, the second step assay, and the second step dilution assay may be performed on the cartridge accommodation part 110 at the same time. However, although only the four cartridges C1 to C4 are illustrated in the present embodiment, more cartridges may be disposed in the stage accommodation part 110 by combining the plurality of cartridges.

(85) Before each reaction starts, a process of removing or punching a cartridge film may be performed. For example, the plurality of cartridges C1 to C4 are disposed in the first to fourth rows M1 to M4 of the cartridge accommodation part 110, and then, a punch part (not shown) punches films of the cartridges C1 to C4, which cover the cartridges, and a plurality of tips 210 are respectively mounted. After such a process, preparation of performing the reaction in each cartridge may be completed, and the four reactions described below may start in each cartridge C1 to C4 at the same time, respectively.

(86) The first step assay using the first cartridge C1 is as follows (see FIG. 10).

(87) A first tip 211 that moves relative to a stage 10 on the first row M1 suctions a sample (step S11) and then sequentially put in empty wells W15 and W14 (step S12) (here, simultaneously, a diluent is suctioned to perform dilution by each of the second tip 212 and the fourth tip 214 in the second cartridge C2 and the fourth cartridge C4), the sample is mixed with a conjugate solution stored in a well W16, and the mixed solution is suctioned again to the first tip 211 (step S13).

(88) Thereafter, the mixed solution in the first tip 211 is discharged into a well W17 in which a fixture solution containing magnetic particles are stored and then mixed and cultured (step S14). Here, a lower end of the well W17 is surrounded by a heating block so that the culture process is

performed at a temperature of about 37 degrees.

(89) Thereafter, the first tip **211** suctions and discharges a cleaning solution stored in wells **W19** to **W112** to perform a cleaning process of removing impurities from the mixed solution except for a conjugate (conjugate of luminant-magnetic particle-biomarker) bonded to the magnetic particles (step **S15**). Particularly, in the cleaning process, the cleaning solution stored in a well **W19** is suctioned into the first tip **211**, and a magnetic force applying part **230** is attached around the first tip **211** to fix the magnetic particles inside the first tip **211**. Then, a driving part **240** is driven to discharge only the impurities of the mixed solution within the first tip **211** together with the cleaning solution from the well **W19**. That is, only the impurities may be removed, but the conjugate bonded to the magnetic particles may remain in the first tip **211** through the cleaning process. This, the cleaning process may be repeated using the cleaning solution stored in the plurality of wells **W19** to **W112**. Thereafter, the first tip **211** suctions the pre-trigger stored in the well **117** (step **S16**) to move to an upper side of a well **W119** disposed at the outermost side of the cartridge. Then, when the solution within the first tip **211** is triggered onto the well **W119**, a luminant is measured by a measurement unit **300** moving to a side surface of the well **W119** (step **S17**). Accordingly, the first step assay by the first tip **211** may be completed.

(90) Thereafter, in order to complete the second step assay and the second step dilution assay of the third tip **213** and the fourth tip **214**, the first tip **211** may stand by in the empty well **W19** (the driving part **240** is not driven after being put in the empty well) (step **S18**) and then may sequentially stand by in the empty wells **W113**, **W114**, **W115**, and **W116** (step **S19**), stand by in the empty well **118**, and stand by in the empty well **S119** (step **S111**).

(91) In order to complete the second step assay and the second step dilution assay, which occur in the third tip **213** and the fourth tip **214**, the above-described steps **S18** to **S111** are steps in which the first tip **211** dependently stands by in the empty well.

(92) The second step dilution assay using the second cartridge **C2** is as follows (see FIG. **11**).

(93) A second tip **212** moving relative to the stage **100** on the second row **M2** suctions a sample (step **S21**) and then suctions a diluent stored in a well **W25** to perform dilution of the sample in an empty well **W24** (step **S22**). Thereafter, the diluted sample is mixed with a conjugate solution stored in a well **W26** and then suctioned again to the first tip **211** (step **S23**).

(94) Thereafter, the diluted sample in the second tip **212** is discharged into the well **W27** in which a fixture solution containing magnetic particles are stored and then mixed and cultured (step **S24**). Likewise, a lower end of the well **W27** is surrounded by a heating block so that the culture process is performed at a temperature of about 37 degrees.

(95) Thereafter, the second tip **212** suctions and discharges a cleaning solution stored in wells **W29** to **W212** to perform a cleaning process of removing impurities from the mixed solution except for a magnetic particle conjugate (step **S25**).

(96) Thereafter, the second tip **212** suctions the pre-trigger stored in the well **217** (step **S26**) to move to a well **W219** disposed at the outermost side of the cartridge. Then, when the solution within the second tip **212** is triggered onto the well **W219**, a luminant is measured by a measurement unit **300** moving to a side surface of the well **W219** (step **S27**). Accordingly, the second step dilution assay by the second tip **211** may be completed.

(97) Thereafter, in order to complete the second step assay and the second step dilution assay of the third tip **213** and the fourth tip **214**, the second tip **212** may stand by in an empty well **W218** (the driving part **240** is not driven after being put in the empty well) (step **S28**) and then may sequentially stand by in the empty wells **W213**, **W214**, **W215**, and **W216** (step **S29**), stand by in the empty well **218**, and stand by in the empty well **S218** (step **S211**).

(98) In order to complete the second step assay and the second step dilution assay, which occur in the third tip **213** and the fourth tip **214**, the above-described **S28** to **S211** are steps in which the second tip **212** dependently stands by in the empty well.

(99) The second step assay using the third cartridge **C4** is as follows (see FIG. **12**).

(100) A third tip **213** moving relative to the stage **100** on the third row **M3** suctions a sample (step **S31**) and sequentially puts into empty wells **W35** and **W34** (step **S32**) and then puts again into an empty well **W36** (step **S33**). (While the third cartridge **C3** is placed in the empty wells **W35** and **W34**, the dilution is performed at the same time after the diluent is suctioned from the second cartridge **C2** and the fourth cartridge **C4**, and while the third cartridge **C4** is paced in the empty well **W36**, the first cartridge **C1** and the second cartridge **C2** are mixed with the conjugate solution).

(101) Thereafter, the simple in the third tip **213** is discharged into a well **W37** in which a fixture solution containing magnetic particles are stored and then mixed and cultured (step **S34**). Here, a lower end of the well **W37** is surrounded by a heating block so that the culture process is performed at a temperature of about 37 degrees.

(102) Thereafter, the third tip **213** suctions and discharges a cleaning solution stored in wells **W39** to **W312** to perform a cleaning process of removing impurities from the mixed solution except for a magnetic particle conjugate (step **S35**).

(103) Thereafter, the third tip **213** is put into the empty well **W317** (step **S36**) and then stands by above a well **W319** (step **S37**) (while the third cartridge **C3** is placed in the empty well **W317**, the first cartridge **C1** and the second cartridge **C2** suction the pre-trigger, and while the third cartridge **C3** stands by above the well **W319**, the solutions within the first cartridge **C1** and the second cartridge **C2** are triggered).

(104) Thereafter, the mixed solution of the third tip **213** is mixed with the conjugate solution stored in the well **W38** and then cultured (step **S38**).

(105) Thereafter, the third tip **213** suctions and discharges a cleaning solution stored in wells **W313** to **W316** to perform a cleaning process of removing impurities from the mixed solution except for a magnetic particle conjugate (step **S39**).

(106) Thereafter, the third tip **213** suctions the pre-trigger stored in the well **318** (step **S310**) to move to a well **W319** disposed at the outermost side of the cartridge. Then, when the mixed solution within the third tip **213** is triggered onto the well **W319**, a luminant is measured by a measurement unit **300** moving to a side surface of the well **W319** (step **S311**).

(107) The second step dilution assay using the fourth cartridge **C4** is as follows (see FIG. **13**).

(108) A fourth tip **214** moving relative to the stage **100** on the fourth row **M4** suctions a sample (step **S41**) and then suctions a diluent stored in a well **W45** to perform dilution of the simple in an empty well **W44** (step **S42**).

(109) Thereafter, the fourth tip **214** is put into the empty well **W46** to stand by in the empty well **W46** (step **S43**).

(110) Thereafter, the simple in the fourth tip **214** is discharged into a well **W47** in which a fixture solution containing magnetic particles are stored and then mixed and cultured (step **S44**). Here, a lower end of the well **W47** is surrounded by a heating block so that the culture process is performed at a temperature of about 37 degrees.

(111) Thereafter, the fourth tip **214** suctions and discharges a cleaning solution stored in wells **W49** to **W412** to perform a cleaning process of removing impurities from the mixed solution except for a magnetic particle conjugate (step **S45**).

(112) Thereafter, the fourth tip **214** is put into the empty well **W417** (step **S46**) and then stands by above a well **W419** (step **S47**) (while the fourth cartridge **C4** is placed in the empty well **W417**, the first cartridge **C1** and the second cartridge **C2** suction the pre-trigger, and while the fourth cartridge **C4** stands by above the well **W419**, the solutions within the first cartridge **C1** and the second cartridge **C2** are triggered).

(113) Thereafter, the mixed solution of the fourth tip **214** is mixed with the conjugate solution stored in the well **W418** and then cultured (step **S48**).

(114) Thereafter, the fourth tip **214** suctions and discharges a cleaning solution stored in wells **W413** to **W416** to perform a cleaning process of removing impurities from the mixed solution

except for a magnetic particle conjugate (step S49).

(115) Thereafter, the fourth tip **214** suctions the pre-trigger stored in the well **418** (step S410) to move to a well **W419** disposed at the outermost side of the cartridge. Then, when the mixed solution within the fourth tip **214** is triggered onto the well **W419**, a luminant is measured by a measurement unit **300** moving to a side surface of the well **W419** (step S411).

(116) For the convenience of explanation, although each of the processes in which the first step assay, the first step dilution assay, the second step assay, and the second step dilution assay respectively occur on the cartridges, has been described, the reactions occur at the same time in the cartridge accommodation part **110** by the tips **210** that move at the same time and are independently adjusted in pressure by the driving part **240**.

(117) Particularly, the first tip **211** to the fourth tip **214** may move at the same time above the wells disposed in the same column N. That is, the above-described steps **S11**, **S12**, **S31**, and **S41** occur at the same time, the above-described steps **S12**, **S22**, **S32**, and **S42** occur at the same time, the above-described steps **S13**, **S23**, **S33**, and **S43** occur at the same time, the above-described steps **S14**, **S24**, **S34**, and **S44** occur at the same time, the above-described steps **S15**, **S25**, **S35**, and **S45** occur at the same time, the above-described steps **S16**, **S26**, **S36**, and **S46** occur at the same time, the above-described steps **S17**, **S27**, **S37**, and **S47** occur at the same time, the above-described steps **S18**, **S28**, **S38**, and **S48** occur at the same time, the above-described steps **S19**, **S29**, **S39**, and **S49** occur at the same time, the above-described steps **S110**, **S210**, **S310**, and **S410** occur at the same time, and the above-described steps **S111**, **S211**, **S311**, and **S411** occur at the same time.

(118) Also, the first cartridge **C1** is disposed in the first row **M1** of the cartridge accommodation part **110**, the second cartridge **C2** is disposed in the second row **M2** of the cartridge accommodation part **110**, the third cartridge **C3** is disposed in the third row **M3** of the cartridge accommodation part **110**, and the fourth cartridge **C4** is disposed in the fourth row **M4** of the cartridge accommodation part unit **110**, but each of the plurality of cartridges **C1** to **C4** may be disposed in any one of the plurality of rows **M1** to **M8** of the cartridge receiving unit **110**. Thus, the order of the arrangement is not limited thereto.

(119) The method in which at least two or more of the above-described first step assay, first step dilution assay, second step assay, and second step dilution assay occur on one stage may be understood as follows.

(120) The step of performing the first step assay may include: a step (step **S11**) in which a first tip suctions a sample; a step (step **S12**) in which the first tip sequentially stands by in empty wells **W15** and **W14**; a step (step **S13**) of suctioning a conjugate solution stored in a well **W16**; a step (step **S14**) of discharging a mixed solution stored in the first tip to a well **W17**, in which a fixture solution containing magnetic particles is stored, so as to be mixed and cultured; a cleaning step (step **S15**) in which the first tip suctions and discharges a cleaning solution stored in wells **W19** to **W112** to remove impurities except for a conjugate bonded to the magnetic particles; a step (step **S16**) in which the first tip suctions a pre-trigger stored in a well **W117**; a step (step **S17**) in which the first tip moves above a well **W119** disposed at the outermost side of the cartridge to inject the solution stored in the first tip onto the well **W119**, thereby measuring the injected solution; a step (step **S18**) in which the first tip stands by in an empty well **W118**; a step (step **S19**) in which the first tip stands by in empty wells **W113**, **W114**, **W115**, and **W116**; a step (step **S110**) in which the first tip stands by in an empty well **118**; and a step (step **S111**) in which the first tip stands by in an empty well **S119**.

(121) The first step dilution assay may include: a step (step **S21**) in which a second tip suctions a sample; a step (step **S22**) in which the second tip suctions a diluent stored in a well **W25** to dilute the sample in an empty well **W24**; a step (step **S23**) of mixing the diluted sample stored in the second tip with a conjugate solution stored in a well **W26** to suction the mixed solution into the second tip; a step (step **S24**) of discharging the mixed solution stored in the second tip to a well **W27**, in which a fixture solution containing magnetic particles is stored, so as to be mixed and

cultured; a cleaning step (step S25) in which the second tip suctions and discharges a cleaning solution stored in wells W29 to W212 to remove impurities from the mixed solution except for a magnetic particle conjugate; a step (step S26) in which the second tip suctions a pre-trigger stored in a well W217; a step (step S27) in which the second tip moves to a well W219 disposed at the outermost side of the cartridge to inject the solution stored in the second tip onto the well W219, thereby measuring the injected solution; a step (step S28) in which the second tip stands by in an empty well W218; a step (step S29) in which the second tip stands by in empty wells W213, W214, W215, and W216; a step (step S210) in which the second tip stands by in an empty well 218; and a step (step S211) in which the second tip stands by in an empty well S218, and

(122) The second step assay may include: a step (step S31) in which a third tip suctions a sample; a step (step S32) in which the third tip sequentially stands by in empty wells W35 and W34; a step (step S33) in which the third tip stands by in an empty well W36; a step (step S34) of discharging the sample stored in the third tip to a well W37, in which a fixture solution containing magnetic particles is stored, so as to be mixed and cultured; a cleaning step (step S35) of suctioning and discharging a cleaning solution stored in wells W39 to W312 to remove impurities from the mixed solution except for a magnetic particle conjugate; a step (step S36) in which the third chip stands by in an empty W317; a step (step S37) in which the third tip stands by above an well W319; a step (step S38) in which the mixed solution of the third tip is mixed with the conjugate solution stored in a well W38 so as to be cultured; a step (step S39) in which the third tip suctions and discharges a cleaning solution stored in wells W313 to W316 to remove impurities from the mixed solution except for a magnetic particle conjugate; a step (step S310) in which the third tip suctions a pre-trigger stored in a well W317; and a step (step S311) in which the third tip moves to a well W319 disposed at the outermost side of the cartridge to inject the mixed solution onto the well W319, thereby measuring the injected solution, and the second step dilution assay includes: a step (step S41) in which a fourth tip suctions a sample; a step (step S42) in which the fourth tip suctions a diluent stored in a well W45 to dilute the sample in an empty well W44; a step (step S43) in which the fourth tip stands by in an empty well W46; a step (step S44) of discharging the diluted sample stored in the second tip to a well W47, in which a fixture solution containing magnetic particles is stored, so as to be mixed and cultured; a cleaning step (step S45) in which the fourth tip suctions and discharges a cleaning solution stored in wells W49 to W412 to remove impurities from the mixed solution except for a magnetic particle conjugate; a step (step S46) in which the fourth tip stands by in an empty well W417; a step (step S47) in which the fourth tip stands by above an empty well W419; a step (step S48) in which a mixed solution of the fourth tip is mixed with a conjugate solution stored in an well W48 so as to be cultured; a step (step S410) in which the fourth tip suctions a pre-trigger stored in a well W418; and a step (step S411) in which the fourth tip moves to a well W419 disposed at the outermost side of the cartridge to inject the mixed solution onto the well W419, thereby measuring the injected solution. Here, at least two reactions of the first step assay, the first step dilution assay, the second step assay, and the second step dilution assay may occur on one stage at the same time. Also, according to an embodiment of the present invention, the second step assay may include: a step in which at least two tips (e.g., a first tip and a third tip) of a plurality of tips suction samples different from each other, respectively; a step in which at least one tip (e.g., the first tip) of the plurality of tips, which suctions the sample, suctions a conjugate solution, and at least the other tip (e.g., the third tip) stands by in an empty well; a step in which the plurality of tips (e.g., the first tip and the third tip), into which the sample is suctioned, discharge a contents stored in the tip to a well containing a fixture solution so as to be mixed and cultured; a cleaning step in which the plurality of tips (e.g., the first tip and the third tip) suctioning the sample suction and discharge a cleaning solution to remove impurities except for a magnetic particle conjugate; a step in which at least one tip (e.g., the first tip) suctioning the conjugate solution suctions a pre-trigger, and at least the other tip (e.g., the third tip) stands by above an empty well; a step in which at least one tip (e.g., the first tip) suctioning the pre-trigger

injects the content contained in the tip onto a well to measure the injected solution by using a measurement unit, and at least the other tip (e.g., the third tip) stands by above an empty well; a step in which at least one tip (e.g., the first tip) injecting the content contained in the tip onto the well stands by in an empty well, and at least the other tip (e.g., the third tip) suctions a conjugate solution; a cleaning step in which at least one tip (e.g., the first tip) injecting the content contained in the tip onto the well stands by in an empty well, and at least the other tip (e.g., the third tip) suctions and discharges the cleaning solution to remove impurities except for a magnetic particle conjugate; a step in which at least one tip (e.g., the first tip) injecting the content contained in the tip onto the well stands by in an empty well, and at least the other tip (e.g., the third tip) suctions the pre-trigger; and a step in which at least one tip (e.g., the first tip) injecting the content contained in the tip onto the well stands by above the well, and at least the other tip (e.g., the third tip) injects the content contained in the tip onto the well to measure the injected solution by using a measurement unit.

(123) Also, a step in which at least one (e.g., the second tip and the fourth tip) of the plurality of tips suctions a diluent before suctioning the conjugate solution and the fixture solution to dilute the sample in the empty well may be further performed.

(124) Also, according to an aspect of the present invention, the immunoassay method may include a step in which at least the other tip stands by above an empty well while at least one tip of a plurality of tips suctions a solution stored in a well or discharges the suctioned solution from the well so that a plurality of reaction methods are performed on one state at the same time.

(125) Also, according to an aspect of the present invention, the plurality of reaction methods may include two or more assays of a first step assay, a first step dilution assay, a second step assay, and a second step dilution assay.

(126) Also, according to an aspect of the present invention, the plurality of reaction methods may include a first step assay and a second step assay, and the immunoassay method may further include: a step in which the tip for performing the second step assay stands by above the empty well while the tip for performing the first step assay suctions a conjugate solution; and a step in which the tip for performing the first step assay stands by above the empty well while the tip for performing the second step assay suctions the conjugate solution.

(127) Also, according to an aspect of the present invention, the immunoassay method may further include: a step in which the tip for performing the second step assay stands by above the empty well while the tip for performing the first step assay suctions a pre-trigger solution; and a step in which the tip for performing the first step assay stands by above the empty well while the tip for performing the second step assay suctions the pre-trigger solution.

(128) Also, according to an aspect of the present invention, the immunoassay method may further include: a step in which the tip for performing the second step assay stands by above the empty well while the solution stored in the tip for performing the first step assay is measured by a measurement unit; and a step in which the tip for performing the first step assay stands by above the empty well while the solution stored in the tip for performing the second step assay is measured by the measurement unit.

(129) Also, according to an aspect of the present invention, at least one assay of the first step assay and the second assay may additionally include a step of diluting a sample, and the immunoassay method may further include a step in which the other tip stands by above the empty well while the tip for performing the first step assay or the tip for performing the second step assay suctions a diluent.

(130) Also, according to an aspect of the present invention, the plurality of reaction methods may include a first step assay and a first step dilution assay, and the immunoassay method may further include: a step in which the tip for performing the first step assay stands by above the empty well while the tip for performing the first step dilution assay suctions the diluent.

(131) Also, according to an aspect of the present invention, the plurality of reaction methods may

include a second step assay and a second step dilution assay, and the immunoassay method may further include: a step in which the tip for performing the second step assay stands by above the empty well while the tip for performing the second step dilution assay suctions the diluent.

(132) Also, according to an aspect of the present invention, a change in pressure within the tip that stands by above the empty well may not occur.

(133) Hereinafter, operations and effects of an immunoassay device and immunoassay method according to an embodiment of the present invention will be described.

(134) The immunoassay device according to the embodiment of the present invention may perform various reaction methods on one stage.

(135) Also, a plurality of cartridges C may respectively include a plurality of reagents and be disposed to be provided with at least two empty wells (each of a first cartridge C1 and a second cartridge C2 includes seven empty wells (except for an empty well for storing a sample), and a third cartridge C3 and a fourth cartridge C4 includes four empty wells), and thus, a plurality of reaction methods may be performed on a stage at the same time.

(136) Also, the plurality of cartridges C may be arranged so that at least one of arrangements of columns of the wells in which a conjugate solution and a pre-trigger solution are stored is different, and arrangements of columns of wells in which a fixture solution is stored are the same, and thus, the plurality of reaction methods may be performed on the stage at the same time.

(137) Also, since a driving part **240** independently applies a pressure to each of tips **210**, the plurality of reaction methods may be smoothly performed on one stage **100**.

(138) Also, since a magnet apply part **230** is provided on a side surface of a tip **210**, magnetic particles may be maintained or fixed in the tip to improve precision of a reaction test.

(139) Also, since the magnet **232** and the tip **210** are fitted with respect to each other, magnetic force applied to the magnetic particles inside the tip **210** may be constantly adjusted.

(140) Also, since a moving measurement unit **300** is provided, reaction and measurement of the sample may occur on one stage to reduce an inspection time.

(141) Also, a well disposed at the outermost side may measure a luminant by the measurement unit **300** that is provided to form a darkroom.

(142) Also, an accurate measurement device for flash type measurement as a small device may be provided.

(143) The device and the immunoassay method according to the embodiment of the present invention have been described above as specific embodiments, but these are merely examples, and thus, the present invention is not limited thereto. It should be construed as having the broadest scope according to the basic idea disclosed herein. Those skilled in the art may combine and replace the disclosed embodiments to implement patterns in a shape that is not described, but this is also within the scope of the present invention. In addition, those skilled in the art may easily change or modify the disclosed embodiments based on the present specification, and it is obvious that such changes or modifications fall within the scope of the present invention.

Claims

1. An immunoassay device comprising: a stage configured to accommodate a plurality of cartridges, each cartridge of the plurality of cartridges having a plurality of wells that are opened upward, wherein the stage is configured to surround a circumference of the well disposed at the outermost side of the cartridges; a solution transfer unit comprising a plurality of tips that are configured to move relative to the stage, are disposed to correspond to positions of the each cartridge, and configured to suction a solution stored in the wells or discharge the suctioned solution from the wells; and a measurement unit disposed at one side of the stage to move in a direction in which the plurality of cartridges are arranged, and comprising a detection unit for measuring an inside state of the well disposed at the outermost side of the each cartridge, and a

shielding plate configured to move to cover an opened upper portion of the well disposed at the outermost side of the each cartridge to block introduction of light into the well, wherein a hole through which an end of a tip of the plurality of tips is introduced into an upper portion of the well disposed at the outermost side of the cartridges, is provided in the shielding plate, the hole of the shielding plate has a size less than that of an upper hole of the well disposed at the outermost side of the cartridges, and when the tip of the plurality of tips is introduced into the well disposed at the outermost side of the cartridges, light incident into the well through a second hole of the shielding plate is blocked.

2. The immunoassay device of claim 1, wherein a first hole for measuring a state of the solution is provided in one side surface of the stage that surrounds the circumference of the well disposed at the outermost side of the cartridges, the detection unit is provided on one side surface of the measurement unit; and when the detection unit is disposed on the side surface of the well disposed at the outermost side of the cartridges to measure the state of the solution, the light incident into the well through the first hole is blocked.

3. An immunoassay device comprising: a stage configured to accommodate a plurality of cartridges, each cartridge of the plurality of cartridges having a plurality of wells that are opened upward, wherein the stage is configured to surround a circumference of the well disposed at the outermost side of the cartridges; a solution transfer unit comprising a plurality of tips that are configured to move relative to the stage, are disposed to correspond to positions of the each cartridge, and configured to suction a solution stored in the wells or discharge the suctioned solution from the wells; and a measurement unit disposed at one side of the stage to move in a direction in which the plurality of cartridges are arranged, and comprising a detection unit for measuring an inside state of the well disposed at the outermost side of the each cartridge, and a shielding plate configured to move to cover an opened upper portion of the well disposed at the outermost side of the each cartridge to block introduction of light into the well, wherein the detection unit and a hole formed in the shielding plate are provided in plurality to measure states of the solutions stored in the plurality of wells disposed at the outermost side of the cartridges at the same time.

4. The immunoassay device of claim 1, wherein the solution transfer unit comprises a magnetic force applying part that is capable of applying magnetic force toward the tip, and when the solution stored in the tip is injected into the well disposed at the outermost side of the cartridges, the magnetic force applying part approaches the tip, and magnetic particles are held inside the tip.

5. The immunoassay device of claim 1, further comprising a measurement unit driving part that is capable of driving the measurement unit in at least one direction, wherein the measurement unit is reciprocated along the wells, which are disposed at the outermost side of the cartridges.
